

UNIVERSITY OF MALTA
FACULTY OF INFORMATION & COMMUNICATION
TECHNOLOGY
DEPARTMENT OF INFORMATION SYSTEMS
CIS3111 Cloud Computing: 2025/26 Assignment on Cloud Security
Secure Cloud Native Computing

Instructions:

1. This is an **individual assignment**.
2. You may be required to demonstrate and present your work to an exam board.
3. A soft-copy of the report, which should include a Google Drive link shared with mark.vella@um.edu.mt containing all the digital artifacts, must be uploaded to the VLE upload area by the deadline below. It is the student's responsibility to ensure that all content is correctly made available for download. The Google Drive content cannot be modified following submission.
4. Assignment submissions require adherence to UM's Responsible Use of Generative AI guidelines¹. Submissions must clearly acknowledge any use of Generative AI for source code and configuration file generation. For any AI-generated content, provide a comprehensive explanation of the generated content along with the full prompting sequence. Ensure that AI usage is transparent and traceable within your submission.
5. Reports that are difficult to follow due to low quality in the writing-style/organisation/presentation will be penalised.
6. Assignment queries are to be made strictly on the assignment's forum on VLE or during tutorials.
7. This assignment is marked out of 100 and comprises 30% of the final CIS3111 assessment.

¹<https://www.um.edu.mt/maks/policiesandprocedures/responsibleuseofgenerativeai/>

8. You are to allocate 25 hours for this assignment.
9. The report must be organized according to the task sequence and for each, concisely explaining/discussing the solution and referring to the associated digital artifacts where relevant.
10. This assignment comprises 3 compulsory tasks.
11. The firm submission deadline is **25th May 2026 at NOON**.

An online book library service, *Books*, is currently hosted on a stand-alone server as shown in Figure 1. It consists of an application server and a PostgreSQL database, each hosted in its own Docker container to simplify deployment.

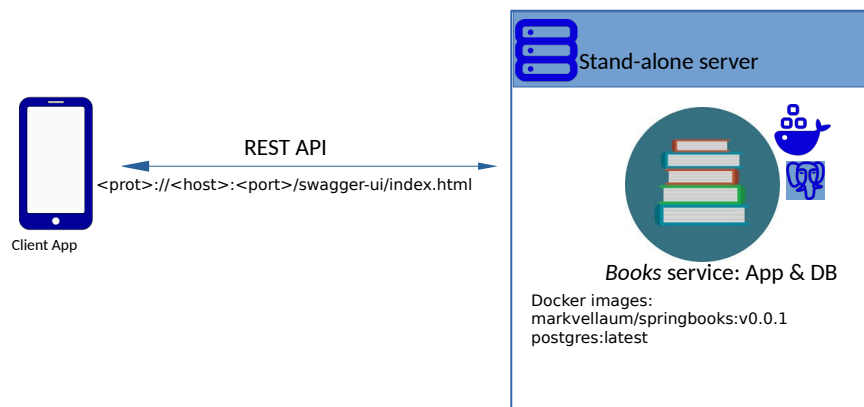


Figure 1: The current *Books* online service hosted on a stand-alone server.

Aiming for high availability, the developers plan to host the *Books* service on a managed Kubernetes platform. The cloud setup, as shown in Figure 2, benefits service discovery, auto-replication, and centralized identity & access management, among other cloud-native architectural benefits.

The development team is currently only in the planning phase of the migration. It is required to replicate aspects of both deployments locally to understand how to proceed.

1. Task 1: Current deployment (Total - 30 marks)

By making use of the official docker documentation², and any other sources as deemed relevant:

- (a) Deploy the *Books* application (markvellaum/springbooks:v0.0.1) and the PostgreSQL database (postgres:latest) inside two separate Docker containers, replicating the deployment in Figure 1. The *Books* application requires the following environment variables: PGHOST, PGPORT, PGDB, PGUSER, PGPASS, for PostgreSQL's

²<https://docs.docker.com/get-started>, <https://docs.docker.com/engine/reference/run/>

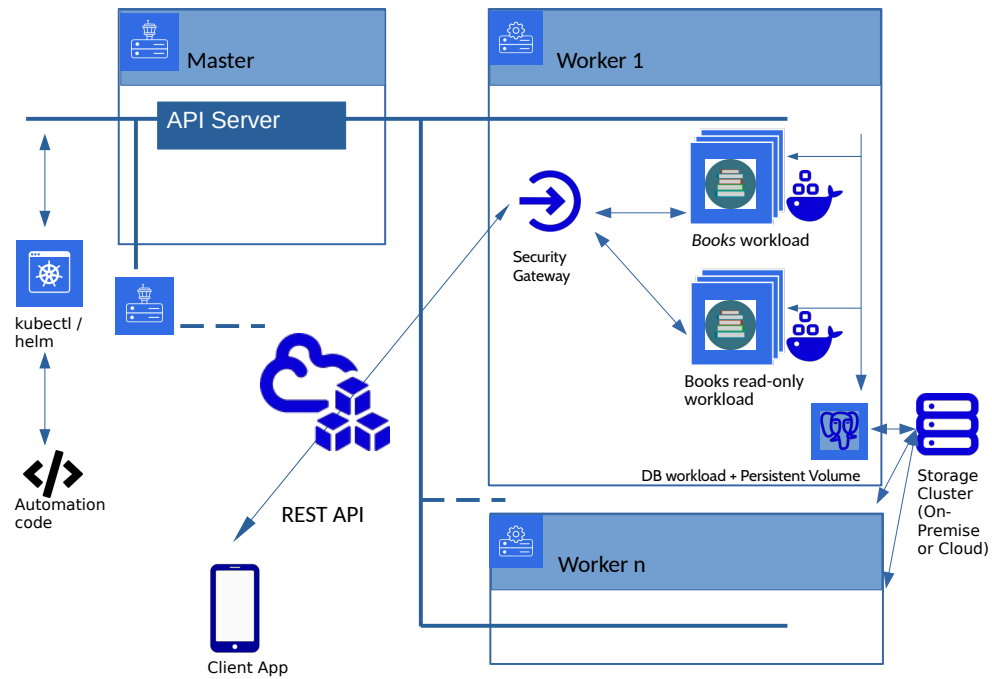


Figure 2: The planned *Books* cloud service deployed on Kubernetes.

host/port/database/user/password, respectively. The PostgreSQL server, by default, listens on port 5432 and accepts the default user `myuser` on installation. Automate the entire setup using any scripting/programming language of your choice.

[15 marks]

- (b) *Books* exposes an OpenAPI documentation of its REST API on the `<prot>://<host>:<port>/swagger-ui/index.html` URI. Import to Postman³ and comprehensively test the endpoints.

[15 marks]

Digital artifacts: Deployment automation script; Test (raw) request/response dumps from Postman; Exported PostgreSQL⁴ table (post-

³<https://www.postman.com/downloads/>

⁴<https://www.postgresql.org/docs/current/sql-commands.html>, <https://www.postgresql.org/docs/current/sql-copy.html>

testing); A `readme.txt` explaining how to replicate the solution.

2. Task 2: Kubernetes deployment (Total - 40 marks)

By making use of lecture notes, the official documentation for Kubernetes⁵ (K8s), minikube⁶ and Helm⁷, and any other sources as deemed relevant: Migrate the deployment from Figure 1 to minikube, to start preparing for, and test, the target deployment shown in Figure 2. Your solution has to take care of the following:

- (a) A Helm installation of PostgreSQL (bitnami/postgresql⁸) to set up a single read/write instance of the database. The database is to be reachable only from other workloads within the cluster. Confirm correct K8s resource creation using `kubectl`.

[10 marks]

- (b) Deploy workloads for markvellaum/springbooks:v0.0.1, the original variant of *Books* intended for librarians, and markvellaum/springbooksro:v0.0.1, a read-only variant of *Books* intended for book borrowers. The expected traffic ratio for the two app variants stands at 1:4. Both are to be reachable by clients outside the cluster.

[10 marks]

- (c) Use `kubectl` to verify that all required Kubernetes resources have been successfully deployed and are functionally operational, along with replication, scaling, and self-healing.

[10 marks]

- (d) Automate the entire deployment and verification setup using any scripting/programming language of your choice.

[10 marks]

Digital artifacts: All configuration files; Deployment automation script; Complete `kubectl` session dumps; Application (pod) logs; A `readme.txt` explaining how to replicate the solution.

⁵<https://kubernetes.io/docs/home/>

⁶<https://minikube.sigs.k8s.io/docs/start/>

⁷<https://helm.sh/docs/>

⁸<https://artifacthub.io/packages/helm/bitnami/postgresql>

3. Task 3: High-availability database cluster (Total - 30 marks)

- (a) In the current workload, both the original and read-only variant of *Books* connect to a single PostgreSQL instance. Replace this configuration with a split Read-write (rw)/Read-only (ro) PostgreSQL cluster using CloudNativePG's (CNPG)⁹ helm charts¹⁰ to deploy a CNPG operator. Apply a manifest YAML to bootstrap an empty cluster¹¹ with a database for the API service. Confirm correct K8s resource creation using `kubectl`.

[12 marks]

- (b) Re-configure the original API service to connect to the rw database and the read-only version to the ro database. Test both versions using Postman.

[10 marks]

- (c) Explain how this setup is more secure in terms of: i) Least Privilege, ii) Traffic spikes, iii) Availability during failures, iv) Data Loss Prevention.

[8 marks]

Digital artifacts: All configuration files; Deployment automation script; Complete `kubectl` session dumps; Test (raw) request/response dumps from Postman and application (pod) logs; A `readme.txt` explaining how to replicate the solution.

⁹<https://cloudnative-pg.io/documentation/1.27/architecture/>

¹⁰<https://github.com/cloudnative-pg/charts/blob/main/charts/cluster/README.md>

¹¹<https://cloudnative-pg.io/documentation/current/bootstrap/#bootstrap-an-empty-cluster-initdb>